



Environmental heterogeneity caused by large-scale cultivation of *Pyropia haitanensis* shapes multi-group biodiversity distribution in coastal areas

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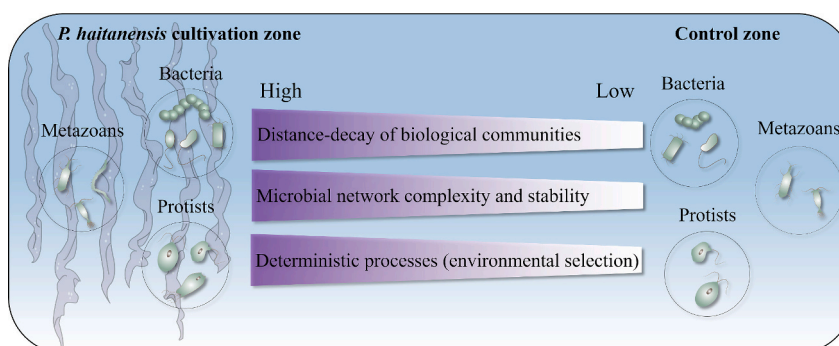
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HIGHLIGHTS

- We compared the communities of bacteria, protists, and metazoans between *P. haitanensis* culture (PZ) and control zones (CZ).
- Three biological communities showed stronger geographic distance-decay patterns in PZ.
- Microbial communities (bacteria and protists) exhibited more complex and stable network structures in PZ.
- Environmental selection was more influential in shaping the three biological communities assembly in the PZ.
- Determinism was the main assembly mechanism of the metazoan community in PZ.

GRAPHICAL ABSTRACT



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ABSTRACT

The response of marine biodiversity to mariculture has long been a research focus in marine ecology. However, the effects of seaweed cultivation on biological community assembly are poorly understood, especially in diverse communities with distinct ecological characteristics. In this study, we used environmental DNA metabarcoding to investigate the spatial distribution patterns of bacterial, protistan, and metazoan diversity, aiming to reveal the mechanisms of community assembly in the *Pyropia haitanensis* cultivation zone along the Fujian coast, China. We found that, compared with the biological communities in control zones, those in *P. haitanensis* cultivation zones exhibited stronger geographic distance-decay patterns and displayed more complex and stable network structures. Deterministic processes (environmental selection) played a more important role in the assembly of bacterial, protistan, and metazoan communities in *P. haitanensis* cultivation zones, especially metazoan communities. Variance partitioning analysis showed that environmental variables made greater contributions to the diversity of the three types of communities within the *P. haitanensis* cultivation zones than in the control zones. Partial least squares path modeling analysis identified nitrate-nitrogen (NO₃-N), pH, particulate organic carbon (POC), and dissolved organic carbon (DOC) as the key environmental variables affecting biodiversity. Overall, the environmental heterogeneity caused by the large-scale cultivation of *P. haitanensis* could be the

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crucial factor influencing the composition and structure of various biological communities. Our results highlight the importance of the responses of multi-group organisms to the cultivation of seaweed, and provide insights into the coexistence patterns of biodiversity at the spatial scale.

1. Introduction

The ecological effects of aquaculture influence the composition and structure of microbial communities (Liu et al., 2022). *Pyropia* is one of the most valuable cultivated macroalgae in the world (Venkatraman and Mehta, 2019). In 2022, the production and cultivation area of Chinese *Pyropia* species reached an impressive 217,658 tons (dry weight) and 66,312 ha, respectively. Notably, *Pyropia haitanensis*, cultivated in the southern coastal regions, accounting for >70 % of the total yield of *Pyropia* species in China (China Fishery Bureau, 2023). Seaweed cultivation not only has an economic value but also provides considerable ecological benefits to the marine ecosystem (Wu et al., 2015; Chen et al., 2020). The cultivation of *P. haitanensis* can increase dissolved oxygen levels, decrease inorganic nutrient levels, and release dissolved organic matter into seawater, such that driving seasonal changes in microbial communities (Xu et al., 2022a). Ahmed et al. (2021) revealed that the cultivation of *Pyropia yezoensis* markedly affects bacterial communities, as the secretion of organic compounds triggers chemical signal transduction and transportation reactions in these communities. Wang et al. (2020) detected consistent differences in microbial communities in seawater between areas with and without *P. haitanensis* cultivation. In addition to microorganisms, zooplankton and phytoplankton play indispensable roles in marine ecosystem stability (Zamora-Terol et al., 2020). Phytoplankton serves as the foundation of the food web and the primary source of energy (Henson et al., 2021). Zooplankton connects phytoplankton and planktonic microorganisms, facilitating energy and substance transfer to higher trophic levels (Ratnarajah et al., 2023). Aquaculture activities have been proven to mediate the assembly of planktonic and sedimentary bacterial communities (Qin et al., 2022). However, our understanding of how multi-trophic biodiversity and community assembly respond to macroalgae cultivation is limited. Thus, it is necessary to more comprehensively evaluate the effects of environmental heterogeneity caused by the cultivation of *P. haitanensis* on marine biodiversity in different coastal areas.

Extensive studies have focused on biogeographic patterns to decipher mechanisms of community assembly (Jin et al., 2022; Kong et al., 2022). The distance-decay relationship is one of the most prevalent biogeographic patterns that describes the decrease in community similarity with increasing geographic distance (Nekola and White, 1999). From a meta-community perspective, deterministic and stochastic processes are the primary factors that determine the strength of distance-decay of community similarity (Vellend et al., 2014). When communities are shaped by stochastic processes, they are randomly influenced by birth, death, and dispersal events (Jetschke, 2002). Conversely, deterministic factors such as environmental filtering and species interactions represent another mechanism of speciation within communities (Wiens et al., 2010). Specifically, environmental selection (homogeneous selection and heterogeneous selection), dispersal (dispersal limitation and homogeneous dispersal), and ecological drift collectively dominate the assembly mechanism of biological communities (Stegen et al., 2013; Zhou and Ning, 2017). Some studies have constructed correlation networks to reveal potential interactions between species and characterize the prospective biotic deterministic processes (Feng et al., 2017; He et al., 2021). More importantly, the relative importance of environmental selection and dispersal limitation in community assembly depends on the ecological characteristics of various organisms (Astorga et al., 2012; Azovsky et al., 2020). For instance, Wang et al. (2012) showed that β diversity is lower for bacterial communities than for communities of eukaryotic microorganisms or macroorganisms because of the wider niche of bacteria and their

stronger dispersal ability, which allow them to adapt to complex and variable environments. Isabwe et al. (2022) also found that bacteria occupy the widest niche and potentially have stronger metabolic plasticity compared with that of larger organisms. Discovering the link between the relative importance of different taxa of organisms and various ecological processes can help to explain the distribution patterns and community assembly mechanisms of biodiversity (Heino et al., 2015) and provide new insights into evaluating the ecological effects of seaweed farming.

Until recently, the study of phytoplankton and microzooplankton in aquatic ecosystems was limited by microscopy methods (Sun et al., 2020; Zhang et al., 2019), which are inefficient and rely on highly specialized taxonomic knowledge (Pawlowski et al., 2018). With the rapid development of high-throughput sequencing technology, environmental DNA metabarcoding (eDNA) has become one of the standard methods used in the field of marine community ecology (Aylagas et al., 2020). The results from eDNA analyses can provide detailed information on the seasonal changes in marine biodiversity across multiple nutrient levels (Djurhuus et al., 2020). The application of eDNA technology at different spatial scales has greatly enhanced the possibilities for biogeographical research (DiBattista et al., 2022). Accordingly, this breakthrough has enabled comprehensive characterization of marine biodiversity across different temporal and spatial scales (Deiner et al., 2017).

In this study, we used eDNA techniques to explore the effects of *P. haitanensis* cultivation on bacterial, protistan, and metazoan diversity along the coast of Fujian, China. Differences in the composition and biogeographical patterns of biological communities were detected by comparisons between two habitats and among three types of organisms. We also connected biogeographical patterns with environmental and spatial variables to decipher the factors affecting bacterial, protistan, and metazoan community assemblies. We hypothesized that the large-scale cultivation of *P. haitanensis* will change the seawater environment, thereby increasing the environmental heterogeneity of the cultivation zones, and that ultimately, these changes will result in different taxonomic groups of organisms exhibiting different spatial distribution patterns under environmental filtering. This study explores the interaction between macroalgae and marine organisms, and the results provide a theoretical basis for evaluating the ecological benefits of macroalgae cultivation.

2. Materials and methods

2.1. Sample collection and environmental data

The sampling sites were located in Xiapu, Nanri Island, Huian, and Zhangpu which represent four mariculture areas located in the coastal area of Fujian Province, China (Fig. 1). In November 2022, 80 seawater samples were collected from the *P. haitanensis* cultivation zones (PZ) and corresponding uncultivated control zones (CZ) in four mariculture areas at a depth of approximately 30 cm. The longitude and latitude of all sample collection sites are shown in Table S1. Seawater samples (volume, 2.5 L) were collected at each sample site in polyethylene containers. The DNA was collected from each sample by filtering 1 L seawater through a nitrocellulose filter membrane (0.22 μ m, Millipore, Billerica, MA, USA). The filter membranes were stored at -80°C until DNA extraction. The water temperature (T), pH, salinity (Sal), and dissolved oxygen (DO) level were measured on-site. The remaining portion of each water sample was used to measure environmental parameters, including chlorophyll *a* (Chl *a*), total nitrogen (TN), total phosphorus

(TP), nitrate-nitrogen ($\text{NO}_3\text{-N}$), phosphate-phosphorus ($\text{PO}_4\text{-P}$), dissolved organic matter (DOM) (including dissolved organic carbon (DOC), dissolved organic nitrogen (DON), and dissolved organic phosphorus (DOP)), as well as particulate organic matter (POM) (including particulate organic carbon (POC) and particulate organic nitrogen (PON)). The measurements were conducted following the protocols described previously (Chen et al., 2020; Xu et al., 2022a).

2.2. DNA extraction, PCR amplification, and sequencing

The biological DNA was extracted from each sample using a PowerWater DNA isolation kit (Qiagen, Hilden, Germany), following the manufacturer's protocols. The quality and quantity of DNA were determined using a NanoDrop 2000 spectrophotometer (Bio-Rad Laboratories Inc. Hercules, CA, USA). Four metabarcoding primers were used to amplify all 80 eDNA samples. The V3-V4 region of the bacteria 16S ribosomal RNA gene was amplified by PCR using the primer pair 341F: ACTCCTACGGGAGGCAG and 806R: GGACTACHVGGGTWTCTAAT (Berg et al., 2012). The V9 region of the 18S ribosomal RNA gene of protists was amplified by PCR using the primer pair 1380F: TCCCTGCCHTTTGTACACAC and 1510R: CCTTCYGCAGGTTACCTAC (Amaral-Zettler et al., 2009). The mCOLintF and jgHCO2198 region of the metazoan mitochondrial cytochrome C oxidase subunit I (COI) gene was amplified by PCR using primer pair mCOLintF: GGWACWGGWTGAACWGTWTAYCCYCC and jgHCO2198: TANACYTCNGGRTGNC-CRAARAAYCA (Leray et al., 2013).

The PCR reactions were performed in triplicate. Each 20- μL reaction mixture consisted of 4 μL 5 $\times$ FastPfu Buffer, 2 μL 2.5 mM dNTPs, 0.8 μL each primer (5 μM), 0.4 μL FastPfu Polymerase, and 10 ng template DNA. The thermal cycling conditions were as follows: initial denaturation at 95 $^{\circ}\text{C}$ for 2 min, followed by 25 cycles at 95 $^{\circ}\text{C}$ for 30 s, 55 $^{\circ}\text{C}$ for 30 s, and 72 $^{\circ}\text{C}$ for 30 s, with final extension at 72 $^{\circ}\text{C}$ for 5 min. Amplicons were extracted from 2 % agarose gels, purified using the AxyPrep DNA Gel Extraction Kit (Axygen Biosciences, Union City, CA,

USA) according to the manufacturer's instructions, and quantified using QuantiFluorTM-ST (Promega, Madison, WI, USA).

2.3. Data preprocessing

Purified PCR products were quantified using a Qubit[®]3.0 fluorometer (Invitrogen, Carlsbad, CA, USA). The pooled DNA product was used to construct a paired-end library following the Illumina genomic DNA library preparation procedure (Illumina, San Diego, CA, USA). The amplicon library was then sequenced in paired-end mode (2×250) on the Illumina NovaSeq platform by the Shanghai BIOZERON Co., Ltd. (Shanghai, China) using standard protocols. The raw reads have been deposited at the NCBI Sequence Read Archive (SRA) database (Accession Number: SRP464701). FLASH (Magoc and Salzberg, 2011) was used to remove sequence fragments with quality scores below 20 and sequence lengths shorter than 150 bp from the raw data. The valid sequences were then used to define operational taxonomic units (OTUs) at a threshold of 97 % similarity, using Usearch (version 7.1 <http://drive5.com/uparse/>). The phylogenetic tree was constructed using FastTree (Price et al., 2009). For bacteria, sequence alignment was performed using the SILVA (version 138.1) database (Yilmaz et al., 2014). For protistan sequences, PR2 (version 4.12.0) database annotations were used (Guillou et al., 2013), and OTUs annotated to other taxa were removed. To obtain more accurate annotations, two COI databases were merged: Midori (Machida et al., 2017) and the Barcode of Life Data System (Ratnasingham and Hebert, 2007). This allowed us to exclude OTUs from non-metazoans. To mitigate the influence of sequencing depth variations on subsequent analysis, standardized sequences were randomly subsampled from each sample using the R package “phyloseq” (McMurdie and Holmes, 2013).

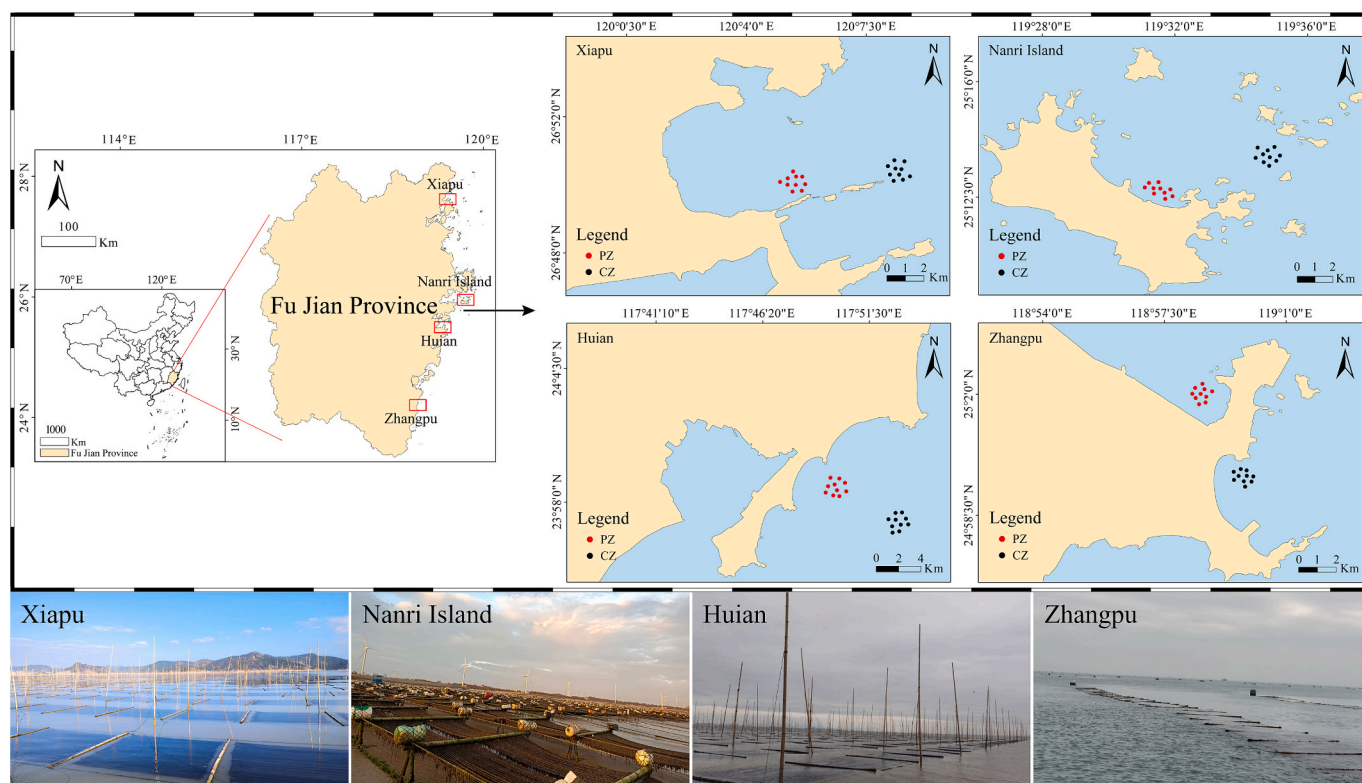


Fig. 1. Map of sampling sites in four mariculture areas (Xiapu, Nanri Island, Huian, and Zhangpu). PZ and CZ represent water samples from *P. haitanensis* cultivation zones and control zones, respectively.

2.4. Statistical analyses

2.4.1. Biodiversity analysis

All analyses were conducted using R (version 4.1.1), and the results were visualized using the “ggplot2” R package (Wickham, 2011). The R package “EasyStat” (<https://github.com/taowenmicro/EasyStat>; v. 0.1.0) was used for statistical testing. The R package “vegan” was used to calculate the Bray-Curtis distance of biological communities and the Euclidean distance of environmental variables. Similarity analysis (ANOSIM) was used to detect differences in biological communities and environmental variables among groups. Principal co-ordinate analysis (PCoA) was conducted based on a Bray-Curtis similarity matrix. The R package “geosphere” was used to calculate the paired geographic distance between the longitude and latitude coordinates of the samples. Then, the distance-decay rate of each biological community was determined by calculating the slope of the ordinary least squares regression between geographic distance and community Bray-Curtis similarity (1 minus Bray-Curtis distance). The Levins' niche breadth index was calculated using the R package “spaa”.

2.4.2. Network analysis

Phylogenetic molecular networks (pMENs) were constructed using the online pipeline iNAP (<https://inap.denglab.org.cn>) (Feng et al., 2022). The OTUs that appeared in 80 % of the samples from different habitats were saved, and after calculating the Spearman's rank correlation matrix, appropriate similarity thresholds were set using random matrix theory (RMT) (Zhou et al., 2010). Networks were visualized using Cytoscape (version 3.9.1) (Shannon et al., 2003). Network properties including power-law model fitness, average clustering coefficient (avgCC), and centralization of betweenness (CB) and connectedness (Con) were calculated. Simultaneously, the topological features at the node level, such as node degree and node closeness centrality, were obtained (Deng et al., 2012). Robustness and vulnerability were assessed to characterize the stability of the networks (Yuan et al., 2021).

2.4.3. Environmental variable analysis

The R package “vegan” was used to calculate the α diversity and β diversity (PCoA1 and PCoA2) of biological communities based on Bray-Curtis distance matrix, and to calculate Spearman's correlations between environmental variables and diversity indexes. We applied multiple regression models and variance decomposition analysis to estimate the contributions of seawater physicochemical properties to explaining the dissimilarities in biological communities (Groemping, 2007). Partial least squares path modeling (PLS-PM) was conducted to analyze the direct and indirect effects of environmental variables on biological diversity and network structures. These analyses were conducted using the R package “plsmp”. Through multiple regression and Spearman's correlation analyses, significant environmental variables were identified. Subsequently, PLS-PM structural equations were constructed, excluding variables with a loading value of <0.7 (Wang et al., 2016). We quantified the relationship between variables using path coefficients. The predictive performance of the model was evaluated using the goodness of fit (GOF) index. The software package “vegan” was used for variation partitioning analysis (VPA) to evaluate the relative importance of environmental and spatial variables, to obtain the principal coordinates of the neighboring matrix (PCNM) of geographical coordinates based on the longitude and latitude of the sampling points, and to calculate the PCNM vectors (Dray et al., 2012). Significant environmental variables and PCNM variables were retained by forward stepwise selection (Blanchet et al., 2008).

2.4.4. Community assembly analysis

The null model was used to evaluate phylogenetic patterns in bacterial, protistan, and metazoan communities through calculating the β Nearest Taxon Index (β NTI) between pairs of samples (Stegen et al., 2012). The R package “picante” was used to calculate the β NTI, which

was estimated by comparing the observed β Mean Nearest Taxon Distance (β MNTD) with the mean of a null distribution of β MNTD (999 randomizations). Subsequently, the β NTI and Raup-Crick Index (RCI) were used to estimate the relative importance of different ecological processes in driving the assembly of different biological communities. Deterministic processes, which include heterogeneous selection and homogeneous selection, were indicated by the proportion of sample pairs with a β NTI absolute value greater than or equal to 2 ($|\beta$ NTI| ≥ 2). Conversely, stochastic processes were recognized when the absolute value of β NTI was lower than 2 ($|\beta$ NTI| < 2). The Bray-Curtis based RCI was calculated, and RCI > 0.95 , RCI < 0.95 , and RCI < -0.95 indicated dispersal limitation, ecological drift, and homogenizing dispersal, respectively.

3. Results

3.1. Composition and diversity of biological communities

In this study, 15,146,022 high-throughput sequences were generated from three DNA libraries from 80 samples, including 5,066,093 for bacteria, 5,047,451 for protists, and 5,032,478 for metazoans. After removing mitochondrial and chloroplast OTUs, 35,926 available OTUs were obtained, comprising 20,859 for bacteria, 6144 for protists, and 8923 for metazoans. The rarefaction curves of all samples tended towards saturation, indicating that the sequencing depth was sufficient to represent almost all bacterial (Fig. S1A), protistan (Fig. S1B), and metazoan (Fig. S1C) communities. The composition of bacterial, protistan, and metazoan communities varied among different habitats. Bacterial communities in all groups were mainly composed of *Proteobacteria* (65.0 % – 74.4 %), *Bacteroidota* (6.0 % – 17.5 %), and *Actinobacteriota* (3.2 % – 10.4 %) at the phylum level (Fig. S2A). The dominant phyla of metazoans in all groups were *Arthropoda* (63.1 % – 84.0 %), *Mollusca* (2.2 % – 13.9 %), *Cnidaria* (2.7 % – 7.7 %), and *Echinodermata* (1.1 % – 10.7 %) (Fig. S2C). The protistan communities in all samples primarily consisted of *Dinophyceae* (7.8 % – 50.6 %), *Cryptophyceae* (3.5 % – 18.4 %), *Ulvophyceae* (0.5 % – 61.9 %), *Coscinodiscophyceae* (3.8 % – 13.9 %), *Mamiellophyceae* (1.7 % – 19.4 %), and *Spirotrichea* (2.5 % – 13.7 %) at the class level (Fig. S2B). Organisms with relatively low abundance were classified as “others”.

The results of the PCoA based on β diversity showed that the different communities exhibited varying extents of separation in space. The first axis explained 26 % – 33 % of the total variance, the second axis explained 17 % – 22 % of the total variance, and the first two axes together explained 43 % – 54 % of the total variance. The ANOSIM results confirmed the significant dissimilarities ($P = 0.001$) in β diversity among the bacterial, protistan, and metazoan communities in both habitats (Fig. 2A, B, and C). Significant ($P < 0.001$) distance-decay relationships were found between the similarity of the three biological communities and geographic distance in different habitats. However, comparing the *P. haitanensis* cultivation zones with control zones, the similarity of bacterial (slope = -5.387 vs. slope = -2.481), protistan (slope = -10.291 vs. slope = -6.059), and metazoan (slope = -4.024 vs. slope = -2.642) communities showed steeper decay slopes in the *P. haitanensis* cultivation zones (Fig. 2D, E, and F). Overall, apart from the significant differences in the β diversity of all three biological groups across the two habitats, the similarity of bacterial, protistan, and metazoan communities showed stronger geographic distance-decay patterns in the *P. haitanensis* cultivation zones than in the control zones.

3.2. Phylogenetic molecular ecological networks of bacterial and protistan communities

The global topological properties of bacterial and protistan networks in the two habitats were calculated (Table S2). All bacterial and protistan empirical networks were well fitted with the power-law model ($R^2 = 0.83$ – 0.98) and exhibited scale-free features, indicating that a few

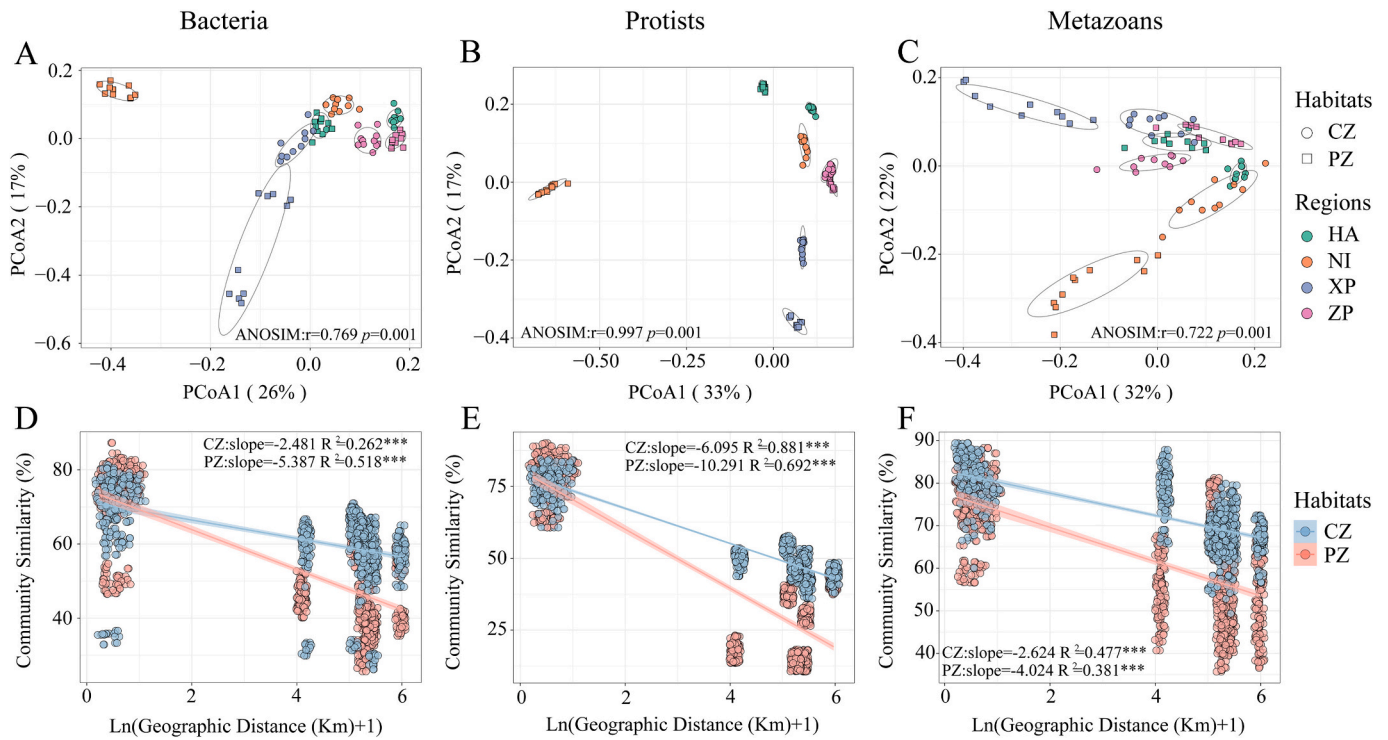


Fig. 2. β diversity and distance-decay relationships of three biological communities in both habitats. Principal co-ordinates analysis (PCoA) of bacterial (A), protistan (B), and metazoan (C) communities and analysis of similarity (ANOSIM) are denoted by the global significance (P -value) and degree of differences (r -value) among all groups. Numbers in parentheses represent percentage of variance explained by each principal coordinate axis. Distance-decay curves show relationships between geographic distance and Bray-Curtis similarity of bacterial (D), protistan (E), and metazoan (F) communities; solid line represents least squares linear regression. Asterisks and R^2 denote significant correlation ($P < 0.001$) and model fit, respectively. PZ and CZ represent water samples from *P. haitanensis* cultivation zones and control zones, respectively.

nodes in the network had numerous connections, while most nodes had either no connections or very few connections. The network of metazoans was not well fitted with the power-law model, so subsequent analyses were not conducted. At the phylum level, bacterial communities were mostly composed of *Proteobacteria*, *Actinobacteriota*, *Bacteroidota*, *Desulfobacterota*, and *Chloroflexi* (Fig. 3A and B) in both habitats. Protist communities mainly consisted of *Dinophyceae*, *Cryptophyceae*, *Coscinodiscophyceae*, *Mamiellophyceae*, and *Spirotrichea* at the class level (Fig. 3C and D) in both habitats. Compared with the control zones, in the *P. haitanensis* cultivation zones, the bacterial (Fig. 3A) and protistan (Fig. 3C) correlation networks showed more interconnected and complex network structures, with higher numbers of edges, nodes, avgCC, CB, and Con (Table S2). Network stability was characterized by calculating the robustness and vulnerability of the network structures. Compared with the network structures in the control zones, those in the *P. haitanensis* cultivation zones manifested significantly higher ($P < 0.001$) robustness and lower vulnerability, indicating higher stability (Fig. 3E and F). Moreover, by obtaining random networks from the empirical networks, the results showed that network indices (e.g. AvgCC, CB and Con) were significantly different ($P < 0.001$) from the respective random networks (Table S2), suggesting that these observed network structures were not randomly connected. That is, the correlation networks between bacterial and protistan communities were more complex and stable in the *P. haitanensis* cultivation zones than in the control zones. This was also confirmed by the significantly higher ($P < 0.01$) topological features at the node level, such as node degree and node betweenness centrality, in the networks of *P. haitanensis* cultivation zones than in the networks of control zones (Table S3).

3.3. Correlations between biological communities and environmental variables

To eliminate collinearity between environmental variables, Spearman's correlation values were calculated for all the environmental variables, and those with a ρ^2 value of < 0.7 were retained for subsequent analysis (Fig. S3). In total, 11 environmental variables were compared in different mariculture areas, and the results are shown in Fig. S4. We found that physical and chemical factors (e.g. temperature (T), pH, and DO), inorganic nutrients (e.g. $\text{NO}_3\text{-N}$, $\text{PO}_4\text{-P}$), dissolved organic matter (e.g. DOC, DON, and DOP), and POC showed varying levels of dissimilarity in both habitats. Euclidean distance was used to evaluate the stability of the environmental factors. The Euclidean distance of environmental variables was significantly higher ($P < 0.001$) in *P. haitanensis* cultivation zones than in control zones (Fig. S5), indicating stronger environmental heterogeneity in *P. haitanensis* cultivation zones. To identify important environmental variables affecting variations in the biodiversity of biological communities, multiple regression models and correlations were used to explore the relationship between individual environmental variables and bacterial, protistan, and metazoan α and β diversity. We found that pH, POC, DOC, and $\text{PO}_4\text{-P}$ were positive predictors for dissimilarities of biological communities in the *P. haitanensis* cultivation zones (Fig. S6). For example, compared with the control zones, in the *P. haitanensis* cultivation zones, pH, POC, and $\text{PO}_4\text{-P}$ were not only significantly related to bacterial and protistan α and β diversity, but were also more important environmental factors. The DOC was also identified as a significant and important environmental variable affecting the β diversity of metazoan communities in *P. haitanensis* cultivation zones.

The PLS-PM analyses further revealed the direct and indirect effects of key environmental variables on biological community diversity and microbial network structures. According to the PLS-PM analyses, POC

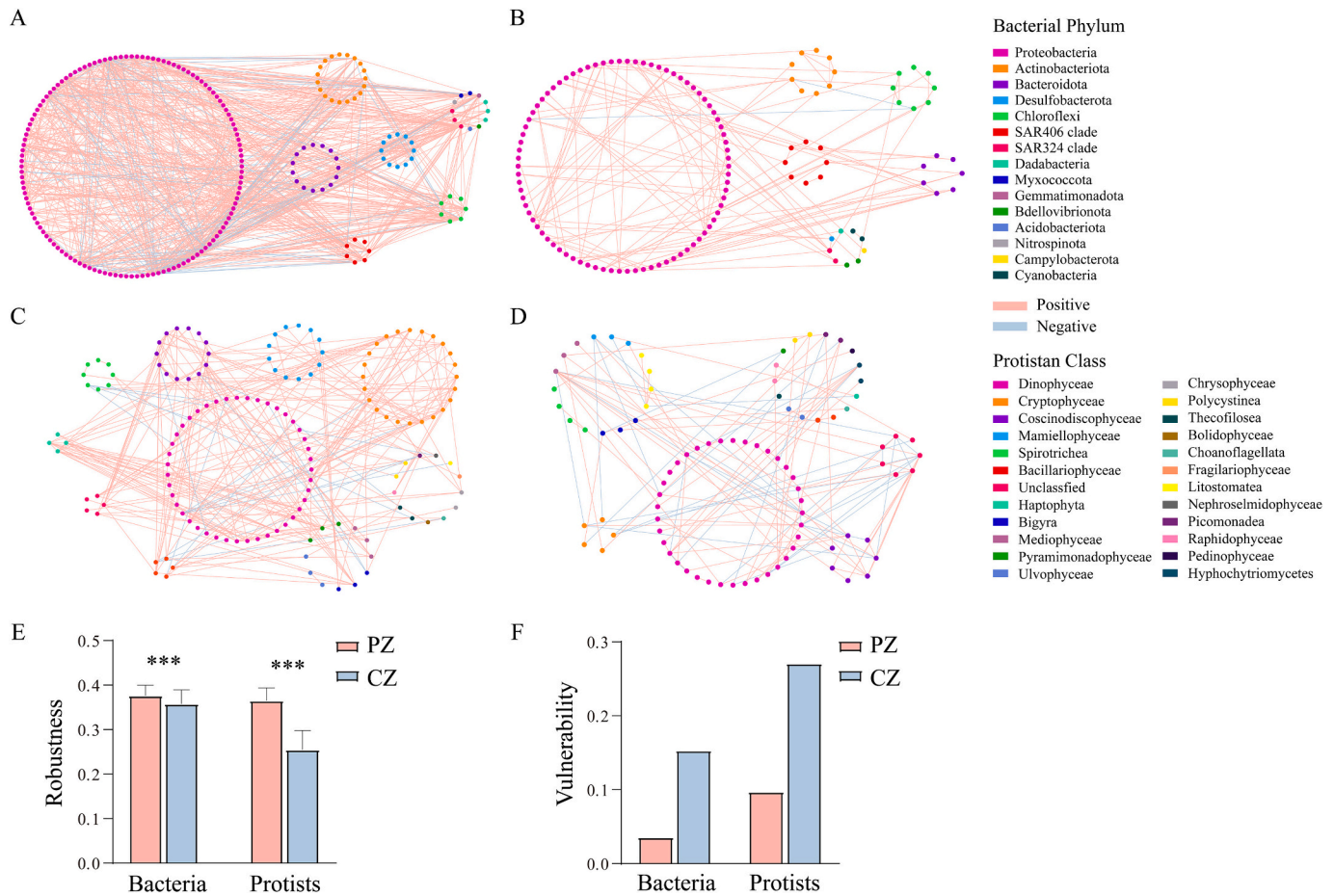


Fig. 3. Correlation networks of bacteria and protists in both habitats. Correlation networks of bacteria (A) and protists (C) in *P. haitanensis* cultivation zones, and correlation networks of bacteria (B) and protists (D) in control zones are indicated by the node color at different taxonomic levels. Positive and negative correlations in the network are indicated by red and blue connections, respectively. Robustness (E) and vulnerability (F) of bacterial and protistan networks in different habitats (calculated as the proportion of taxa remaining after removing 50 % of randomly selected taxa from each empirical network). Error bars represent the standard deviation of 1000 repetitions of the simulation. Asterisks indicate significant difference in robustness of networks between *P. haitanensis* cultivation zones (PZ) and control zones (CZ) (significance level: *** $P < 0.01$, Student's *t*-test).

and pH had significant positive effects ($P < 0.001$) on the diversity of bacterial and protistan communities, respectively, in the *P. haitanensis* cultivation zones (Fig. 4A). Furthermore, $\text{NO}_3\text{-N}$ was identified as a key environmental variable significantly ($P < 0.001$) and positively affecting the diversity of protistan communities, while DOC had a significant positive effect ($P < 0.001$) on the diversity of metazoan communities (Fig. 4A). In contrast, in the control zones, DOC, POC, and $\text{NO}_3\text{-N}$ were the main environmental variables influencing bacterial community diversity, and pH contributed to the diversity of the metazoan community (Fig. 4C). In summary, pH, DOC, POC, and $\text{NO}_3\text{-N}$ were identified as the main factors driving biodiversity in *P. haitanensis* cultivation zones.

3.4. Ecological assembly processes of biological communities

The null model was used to investigate the mechanisms driving the assembly of biological communities. The βNTI and RCI were calculated to quantify the contributions of deterministic processes (including homogeneous selection and heterogeneous selection) and stochastic processes (including dispersal limitation, homogeneous dispersal, and ecological drift) to the assembly of biological communities in both habitats. Comparing the *P. haitanensis* cultivation zones with the control zones, environmental selection (including homogeneous selection and heterogeneous selection) had a higher relative importance for the assembly of bacterial (39.74 % vs. 35.9 %) (Fig. 5A and D), protistan (28.85 % vs. 16.54 %) (Fig. 5B and E), and metazoan (60.38 % vs. 38.33

%) (Fig. 5C and F) communities in the *P. haitanensis* cultivation zones. Notably, the assembly of metazoan communities in the *P. haitanensis* cultivation zones was primarily governed by deterministic processes ($|\beta\text{NTI}| \geq 2$), whereas it was dominated by stochastic processes in the control zones (Fig. 5I). The assembly of bacterial and protistan communities was governed by stochastic processes in both habitats ($|\beta\text{NTI}| < 2$) (Fig. 5G and H).

The results of the VPA showed that environmental factors made a greater contribution to the variations in community composition structure of bacteria (25.6 % vs. 8.2 %), protists (37.9 % vs. 18.3 %) and metazoans (22.3 % vs. 7.2 %) in the *P. haitanensis* cultivation zones than in the control zones (Fig. 6A). This further confirmed the greater importance of deterministic processes in driving biodiversity in *P. haitanensis* cultivation zones than in control zones. The niche breadths differed significantly ($P < 0.001$) among bacteria, protists, and metazoans in both habitats, with metazoans occupying the narrowest niche among the three organisms (Fig. 6B).

4. Discussion

4.1. Key environmental variables affecting the structure of different biological communities

In this study, we discovered that the levels of DOC and POC were generally higher in the *P. haitanensis* cultivation zones than in the

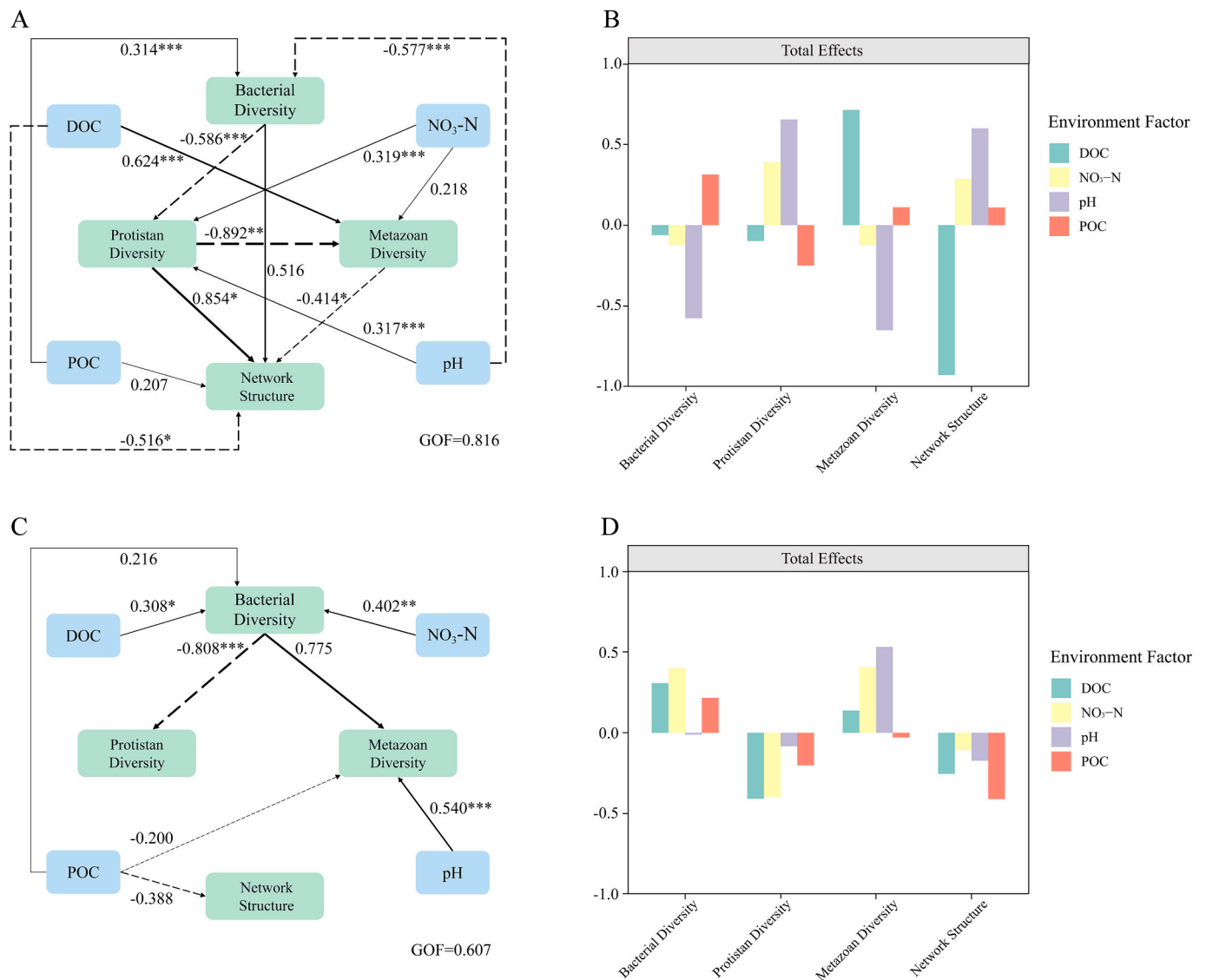


Fig. 4. Partial least squares path models (PLS-PM) analysis of effects of environmental variables on the diversity of bacteria, protists, and metazoans, and on the network structure of bacteria and protists in both habitats. Path structure (A) and total effect value of environmental variables (B) in *P. haitanensis* cultivation zones; and path structure (C) and total effect value of environmental variables (D) in control zones. For simplicity, only paths with coefficient > 0.2 are displayed. Solid lines indicate positive effects, dashed lines represent negative effects. Asterisks indicate significance levels of path coefficients as follows: * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$. Goodness of fit (GOF) index was used to assess model fit.

uncultivated control zones (Fig. S4), which could be mainly attributed to the active release of oxygen and particulate carbon by *P. haitanensis* during the course of its normal physiological functions (Xu et al., 2022b; Xu et al., 2021). The carbon cycle in marine ecosystems is intricately linked to various trophic organisms in the food web (Calbet and Landry, 2004), with POC generated through phytoplankton photosynthesis being sequentially transferred from primary producers to higher trophic levels via the biological carbon pump (Basu and Mackey, 2018). The biodiversity in the *P. haitanensis* cultivation zones was closely correlated with both DOC and POC, and these factors showed different levels of relative importance in shaping the diversity of biological communities (Fig. S6). Importantly, the PLS-PM analysis further revealed significant positive effects of POC and DOC on bacterial and metazoan diversity, respectively (Fig. 4A). Marine metazoans rely on the absorption of food through their digestive wall to fuel metabolism and growth, and this explains their dependence on DOC (Steinberg and Landry, 2017). Bacteria play pivotal roles in the carbon cycle by secreting hydrolytic enzymes to break down POC into soluble POC and DOC (Kong et al., 2021). Our results are consistent with those of a recent study (Liang et al.,

2022), which concluded that the release of DOC by cultivated laver (*Pyropia* sp.) was an important environmental factor influencing bacterial community diversity. These findings highlight the potential of cultivated macroalgae to modify marine community structures and contribute to carbon sequestration in marine ecosystems (Li et al., 2022; Powers et al., 2019).

Moreover, a growing body of evidence suggests that macroalgae such as *Undaria pinnatifida*, *Sargassum fusiforme*, and *Gracilaria lemaneiformis* can alter the physical and chemical properties of seawater through photosynthesis. Studies have shown that these seaweeds affect the pH level of seawater and the concentrations of inorganic nutrients and dissolved oxygen, thus enhancing environmental heterogeneity and driving variations in the composition and structure of microbial communities (Zhang et al., 2021; Zhao et al., 2018; Xie et al., 2017). In the present study, we also found that inorganic nutrients such as NO₃-N and PO₄-P, as well as physicochemical factors like pH and DO, were strongly correlated with biodiversity in *P. haitanensis* cultivation zones (Fig. S6). Furthermore, NO₃-N and pH had a significant positive effect on protistan communities (Fig. 4A). The absence of similar pathway patterns in the

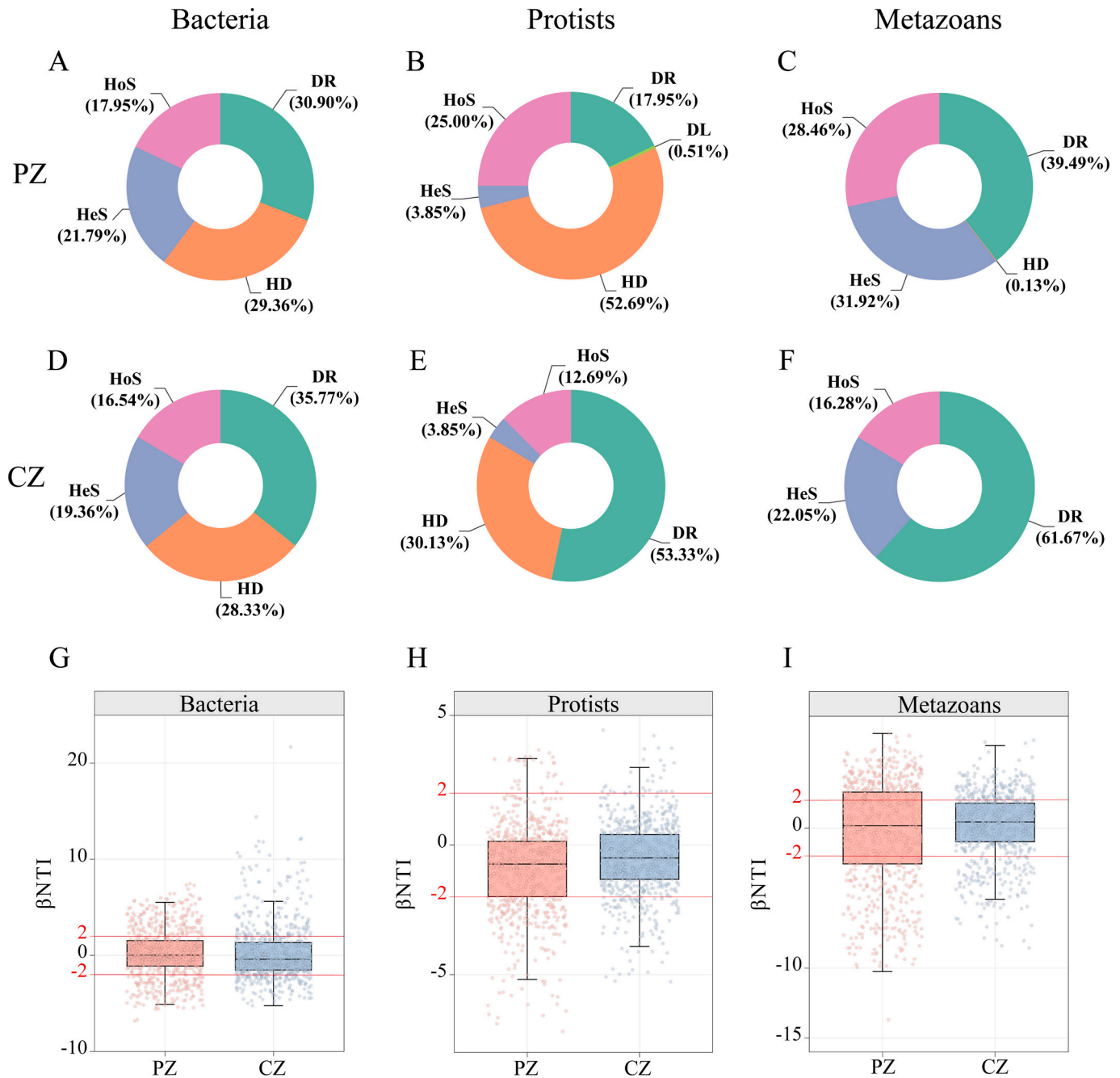


Fig. 5. Null model analysis of three biological communities in both habitats. Figure shows the relative importance of different ecological processes, namely homogeneous selection (HoS), heterogeneous selection (HeS), homogeneous dispersal (HD), dispersal limitation (DL), and drift (DR) to the assembly of bacterial (A), protistan (B), and metazoan (C) communities in *P. haitanensis* cultivation zones, as well as to the assembly of bacterial (D), protistan (E), and metazoan (F) communities in control zones. Contributions of deterministic processes ($|\beta_{NTI}| \geq 2$) and stochastic processes ($|\beta_{NTI}| < 2$) to community assembly of bacteria (G), protists (H), and metazoans (I). PZ and CZ represent water samples from *P. haitanensis* cultivation zones and control zones, respectively.

control zones further emphasized the significant effect of these environmental variables on biological communities in the *P. haitanensis* cultivation zones (Fig. 4C). Photosynthesizing macroalgae absorb dissolved nutrients from seawater, thereby competing with other phytoplankton for nutrients (Wang et al., 2007). For another perspective, different protist species show different degrees of tolerance to changes in pH (Hansen, 2002; Hinga, 2002), this could result in variations in protistan community diversity in *P. haitanensis* cultivation zones. In summary, the variations in the composition and structure of biological communities could be attributed to the combined effects of multiple environmental variables influenced by large-scale *P. haitanensis*

cultivation.

4.2. Structural characteristics of bacterial and protistan correlation networks

Previous studies have identified biological interactions as biotic drivers that influence community composition and structure, providing critical insights into the mechanisms of biological community assembly (Calcagno et al., 2017; Hunt and Ward, 2015). Our results show that, compared with the network structures of control zones, those of *P. haitanensis* cultivation zones exhibited greater complexity, as

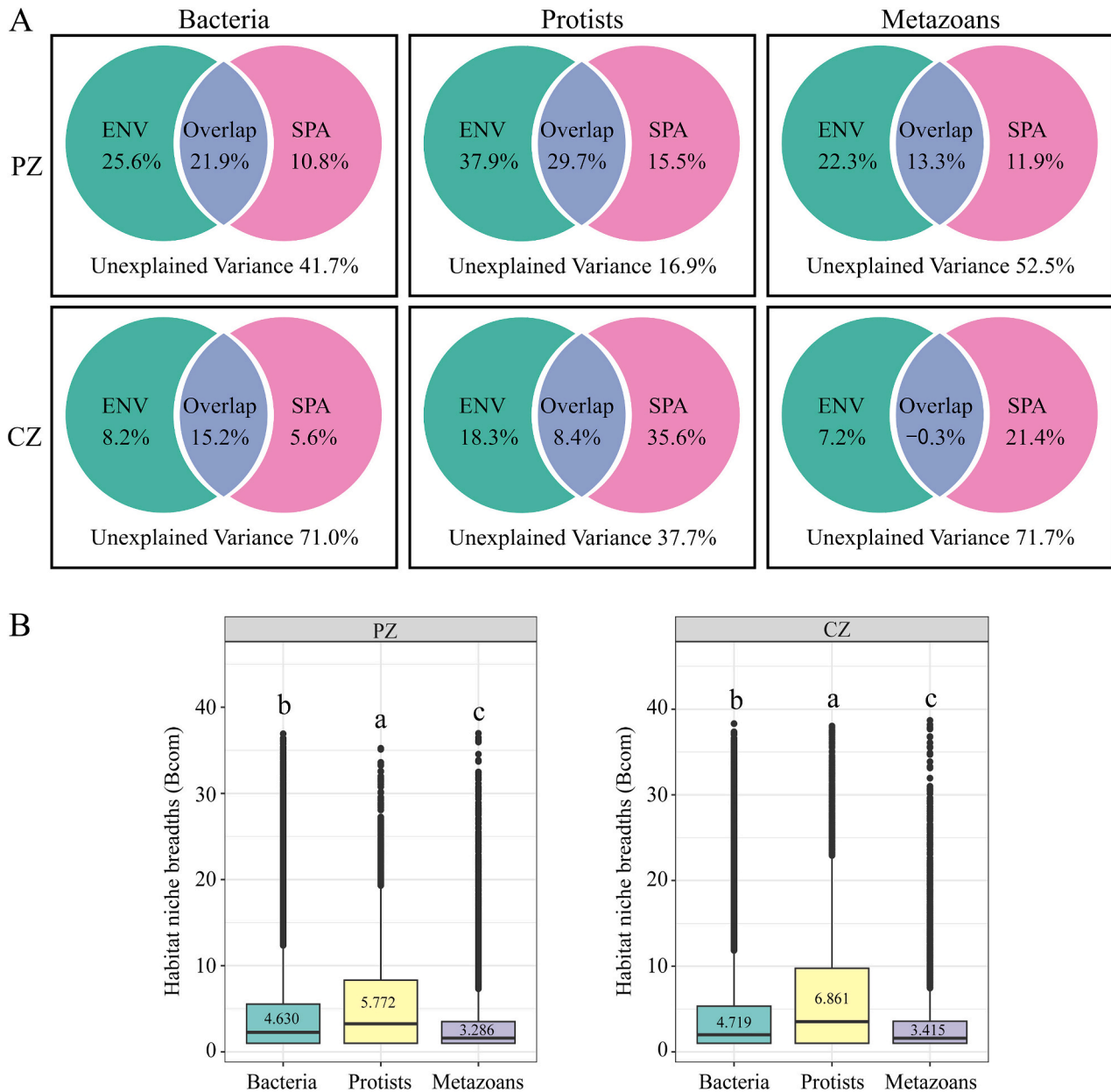


Fig. 6. (A) Variational partitioning analysis to determine the contributions of environmental variables (ENV) and spatial variables (SPA) to the composition structure of bacterial, protistan, and metazoan communities in *P. haitanensis* cultivation zones (PZ) and control zones (CZ). Environmental variables and spatial variables jointly explaining composition structure of communities are represented by overlaps. (B) Mean habitat niche breadths (Bcom) in bacterial, protistan, and metazoan community samples in *P. haitanensis* cultivation zones (PZ) and control zones (CZ) (significance level: $P < 0.001$, Kruskal-Wallis test). Numbers in the box plot are average values.

evidenced by higher scores for nodes, edges, avgCC, CB, and Con (Table S2). These results suggested that the organisms were more interconnected in *P. haitanensis* cultivation zones than in control zones. Previous studies have demonstrated strong correlations between the complexity and stability of microbial correlation networks (Yuan et al., 2021). The high connectivity and complexity of microbial network structures could enhance the efficiency of resource and information transmission, consequently leading to an improvement in the stability of community functions (Morriën et al., 2017). In this study, compared with the control zones, in the *P. haitanensis* cultivation zones the network structures of bacteria and protists showed significantly higher ($P < 0.001$) robustness and lower vulnerability (Fig. 3E and F), indicating that microorganisms within the *P. haitanensis* cultivation zones had more stable network structures. There is evidence that the

complexity of microbial networks contributes substantially to the multifunctionality of ecosystems, and may even determine the trends in, and intensity of, the relationship between biodiversity and function (Maynard et al., 2017). Accordingly, complex and stable microbial network structures may play a non-negligible role in maintaining the functional diversity of the ecosystem in the *P. haitanensis* cultivation zones. Moreover, microbial networks can respond to different levels of disturbance caused by environmental factors (Chen et al., 2023). The results of the PLS-PM analysis showed that $\text{NO}_3\text{-N}$, pH, and POC had direct or indirect effects on the structure of the microbial network in the *P. haitanensis* cultivation zones (Fig. 4B), indicating that the environmental heterogeneity caused by *P. haitanensis* cultivation might possibly be a driver of microbial network structure. Overall, compared with the microbial networks in the control zones, those in the *P. haitanensis*

cultivation zones displayed more complex and stable coexistence patterns. The assembly of microbial communities in the *P. haitanensis* cultivation zones may be influenced by additional biotic deterministic processes.

4.3. Mechanisms shaping the assembly of different biological communities

The concept of “Everything is everywhere, but, the environment selects” emphasizes that environmental factors play a significant role in shaping biodiversity. This concept has been supported by the results of biogeographical studies in small-scale marine environments (Allen et al., 2020; De Wit and Bouvier, 2006; Zhou et al., 2021). Here, we found that the similarity of the three biological communities exhibited distinct geographic distribution patterns in both habitats. Compared with the biological communities in control zones, those in the *P. haitanensis* cultivation zones exhibited steeper decay slopes as the geographic distance increased, suggesting that the similarity of the biological communities in the cultivation zones followed stronger geographic distance-decay patterns (Fig. 2D, E, and F). From the perspective of specific ecological processes, the extent of decay between geographic distance and community similarity is mainly influenced by environmental selection and dispersal limitation (Zhang et al., 2018). Consistent with our expectations, we found that environmental selection (including homogeneous selection and heterogeneous selection) showed higher relative importance for the assembly of the three biological communities in the *P. haitanensis* cultivation zones (Fig. 5 A, B, and C) than in the control zones (Fig. 5 D, E, and F). The VPA results further confirmed that environmental variables made greater contributions to the composition of biological communities in the *P. haitanensis* cultivation zones than in the control zones (Fig. 6A). The Euclidean distance of environmental variables was significantly higher ($P < 0.001$) in the *P. haitanensis* cultivation zones than in the control zones (Fig. S5), although significant differences in not all environmental variables were detected in both habitats (Fig. S4). These results revealed that the heterogeneity of the environment was greater in the *P. haitanensis* cultivation zones than in the control zones. Consequently, the differences in the distance-decay relationships of the similarity of the three biological communities between the two habitats could be partly attributed to large-scale *P. haitanensis* cultivation, which enhanced environmental heterogeneity to support a wider range of biological preferences.

A recent study found that, apart from being influenced by environmental heterogeneity, the relative importance of environmental selection and dispersal limitation in community assembly is also affected by different types of organisms (Zhao et al., 2021). Interestingly, we observed apparent differences in the dominant ecological processes governing the assembly of different biological communities in the *P. haitanensis* cultivation zones. Stochastic processes mainly governed the assembly of bacterial and protistan communities (Fig. 5 G and H), while deterministic processes governed the assembly of the metazoan community (Fig. 5I). In contrast, all three types of communities were dominated by stochastic processes in the control zones (Fig. 5 G, H, and I). These findings support the ecological hypothesis of “size plasticity,” which assumes that larger organisms occupy narrower niches, and their metabolic rate is less plastic, so they are more susceptible to environmental filtering than are smaller organisms (Langenheder and Ragnarsson, 2007). The niche breadths of different organisms were calculated to test this hypothesis. Significant differences ($P < 0.001$) were consistently observed in the niche breadths of bacteria, protists, and metazoans in both habitats. Metazoans occupied the narrowest niche among the three biological communities (Fig. 6B). Accordingly, metazoans were influenced by stronger environmental filtering in *P. haitanensis* cultivation zones, which may be the critical reason why determinism was the main mechanism affecting metazoan community assembly. The “size plasticity” hypothesis is also supported by the results of a study on the geographic distribution of biological communities in the East China Sea, where smaller organisms (bacteria) were found to

be less affected by environmental filtering than were larger organisms (protists) (Wu et al., 2018). The size of organisms affects their strengths in shaping global geographic distribution patterns (Azovsky et al., 2020). Further comparative studies should be conducted on a larger spatial scale to explore the assembly mechanisms of different types of biological communities in *P. haitanensis* cultivation zones and to reveal the broadly applicable mechanisms that shape community assemblies.

5. Conclusion

We used eDNA to identify distinct geographic distribution patterns of three biological communities in two habitats: *P. haitanensis* cultivation zones and control (uncultivated) zones. Compared with the control zones, in the *P. haitanensis* cultivation zones, the similarity of bacterial, protistan, and metazoan communities exhibited stronger distance-decay patterns. Deterministic processes (environmental selection) were more important for the assembly of bacterial, protistan, and metazoan communities in the *P. haitanensis* cultivation zones than in the control zones because of the greater environmental heterogeneity in the *P. haitanensis* cultivation zones. This was especially pronounced for metazoan communities because of their narrow niche. Compared with bacterial and protistan communities in the control zones, those in the *P. haitanensis* cultivation zones displayed more complex and stable co-occurrence network structures, potentially influencing their community assembly. Our findings emphasize the significance of biological characteristics in community assembly and highlight the potential effects of large-scale *P. haitanensis* cultivation on offshore biological communities. Future studies should expand the spatial scale to better understand the ecological effects of large-scale seaweed cultivation.

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CRedit authorship contribution statement

Zongtang Li: Writing – original draft, Visualization, Validation, Formal analysis, Data curation. **Kai Xu:** Supervision, Resources, Project administration, Investigation. **Muhan Meng:** Resources, Investigation. **Yan Xu:** Supervision, Project administration. **Dehua Ji:** Supervision, Project administration. **Wenlei Wang:** Writing – review & editing, Funding acquisition. **Chaotian Xie:** Writing – review & editing, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare no conflicts of interest.

Data availability

The raw sequence data reported in this paper have been deposited in the National Center for Biotechnology Information's Short Reads Archive (SRA) database under the BioProject number PRJNA1022594.

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